

Smart Email Management Project Proposal Report

CO3554 Data Management Project

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Introduction

The digital age has made e mail services play an essential role in professional or social communications. Nonetheless, the sheer volume of daily communication via this method may potentially result in information overload-related difficulties that may complicate the management of electronic mail services, and become an obstacle to efficient productivity. Though electronic mail client applications have basic capabilities in electronic mail classification, like filtering out spams, the unavailability of an efficient system that may perform intelligent content-based sorting of electronic mail content based on positive/negative sentiments become essential reasons in creating an efficient system like this.

The proposed project proposes a way of solving this major problem through a combination of data mining on emails and machine learning principles to come up with a smart way of managing emails. Through publicly available data repositories like Enron Email Dataset and SpamAssassin dataset, a system would be developed to categorize emails under specific classes like work-related, personalized, spam messages, and urgent messages. Furthermore, there would be a way of understanding sentiment from emails and ensuring that a user would be in a position to know what action to take on messages from others. An analytics dashboard will allow users to visualize communication patterns and track the effectiveness of the email classification system.

The importance of this project lies in the fact that it extends beyond traditional methods of deciding which emails to filter out and provides users with a transparent approach to email management through explainability and data-driven methods. The problem solved in this project is a real-world scenario to which both users and companies can use to increase their productivity and avoid overlooking important emails and enable making important decisions based on this system. Additionally, through various machine learning model implementations in this project, it provides users with important insights regarding real-world applications of data mining in text classification.

Problem Statement & Objectives

Problem Statement

While email is increasingly used as a means of personal and professional communication, most users struggle to cope with mass volumes of messages efficiently. Basic filtering provided by already existing email clients, like spam detection falls short of the necessary requirements. A data-driven, intelligent method where emails are classified by content, sentiment, and urgency is needed. Users might miss important messages, waste a lot of time organizing their inbox, or not respond to certain e mail with the correct urgency. There is an expressed need for a system that could intelligently classify, prioritize, and provide actionable insights from emails does all of this while having transparency and being able to adapt for different users and their preferences

Objectives

- Obtain Emails from public available datasets like Enron Dataset and make them ready for data mining by preprocessing them.
- Extract important features from the content and metadata of the emails for the classification, sentiment analysis, and priority prediction.
- Use machine learning to classify emails by type, for instance, being for work, personal use, spam, and urgent.
- Analyze the sentiment and urgency of the email to provide suggestions for the user's next actions.
- Evaluate the model's performance against other baseline models by using basic parameters such as accuracy, precision, recall, and F1-score.
- Design an analytics visual component to display mail patterns, user interactions, and model outputs including an inbox.
- Provide privacy and security by client-side classification with the option to personalize the model by using the server to retrain the models based on the features.

Literature Review

Emails and email classification/prioritization have always been an area of active research in data mining and natural language processing, mainly because of the increasing usage of email services and information overload. Relevant studies have examined the usage of machine learning models in email prioritization based on the emotional content of an email. During this investigation, the effectiveness of different algorithms like Naive Bayes, Support Vector Machines, Logistic Regression, and Random Forest were evaluated over a large labeled dataset to prioritize the emails. During this investigation, it was clearly stated that email classification and prioritization using machine learning models were quite efficient in order to prioritize the emails considering the emotional content. Also, it was clearly evaluated that models like Logistic Regression and Support Vector-based models were quite effective when compared to other models. [1]

Another line of work focuses on exploiting sentiment detection and text semantics for mail processing. To illustrate this, studies on email sentiment classification using advanced models like bidirectional LSTM along with topic embeddings have revealed some promising findings about enhancing mail sentiment detection using semantic and topic-level features in email texts. Accordingly, this tackles issues with mail content dealing with multiple topics or having variable length messages, as these are complex features not easily accounted for using simple features like word presence or absence vectors.[2][3]

Nevertheless, there are still some issues in more thoroughly integrating classification, sentiment analysis, urgency recognition, and priority identification in a single system. As seen in existing works, there are mostly specific focused solution implementations based on each feature, but none in combination with user experience and analysis. Therefore, there are clear enhancements in integrating all these features in a single system through a more holistic approach in email management.

Dataset Overview

The project will be conducted based on two freely available datasets on emails. The Enron Email Dataset and the SpamAssassin Public Corpus dataset. The datasets have real-world data and are suitable for training and test datasets in the development and evaluation of multiple classifications and prioritization tasks on emails.[4][5]

The Enron Email Dataset is a collection of around 500,000 emails sent between employees at Enron. The dataset contains with useful metadata like sender, receiver, time stamps, subject , as well as the body of the email. The main usage of this dataset is to model real-world email behavior for legitimate users and then classify those emails under potential categories like personal emails, business-related emails, and other emails. The dataset looks authentic for purposes like feature extraction, sentiment analysis, priority prediction, etc.

SpamAssassin Dataset: A pre-processed labeled dataset of spam and nonspam email messages. The goal is the generation of spam detection components through training and validation to handle unwanted or malicious emails.

After preprocessing, all the email data will be stored in a structured database. Raw emails will be cleaned, tokenized, and transformed into numerical representations by using text vectorization. Metadata and derived attributes include sentiment scores, predicted class labels, and priority levels, which will also be kept separately for analysis and dashboard visualization. The datasets will be maintained in a versioned format to support reproducibility, model retraining, and evaluation over time.

Proposed Architecture

The proposed system will use a modular and layered architecture for secure email processing, machine learning-based analysis, and user interaction. Four main layers will constitute the architecture. These are Data Ingestion Layer, Processing and Analytics Layer, Model Layer, and Application Layer.

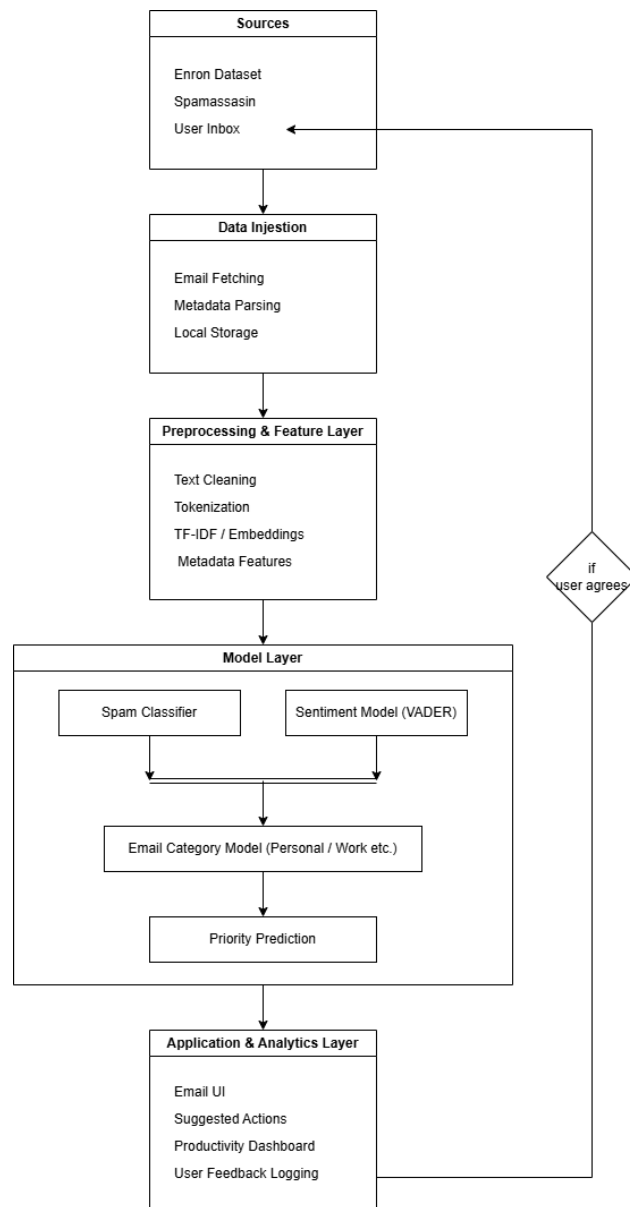


Figure 1: Proposed System Architecture

At the Data ingestion layer, the emails will be obtained from publicly available sources like Enron or SpamAssassin for the development of the model. Optionally they may be retrieved from the inbox for the purpose of the retraining based on the user preference. Emails contain sender details, recipient details, the time stamp for each one of them, the subject heading, the email content itself, etc., For privacy, email content is processed locally during classification.

The Processing Analytics Layer represents the layer responsible for text preprocessing like cleaning, tokenization, stop-word removal, and feature extraction. Structural features can be joined, including metadata such as frequency of sender and time patterns, with the text-based features to form structured inputs into the models.

The Model Layer consists of multiple independent machine learning components that contribute a joint output. This includes a sentiment analysis module, which would label content with a sentiment score; classification models, which categorize emails into spam, personal, or work-related classes; and a priority prediction module that would use the classification results along with the sentiment scores and metadata to estimate email urgency. All the base models are trained offline using the Enron and SpamAssassin datasets. Optional userspecific retraining is done server-side, while anonymizing feedback.

The Application Layer contains a user interface for the email services and an analytics dashboard for displaying user interactions, classified emails, and suggested actions, which are then used to evaluate the user and improve the model, through well-designed APIs for inter-layer communication.

Tools and Technologies

Programming Data Processing

Python is used as the primary programming language because of its comprehensive ecosystem for data management and machine learning. Pandas for efficient cleaning, transformation, and storage of email metadata and derived features. NLTK will be used to tokenize text, remove stop-words, and extract linguistic features.

Machine Learning & Natural Language Processing

Random Forest and XGBoost will be used because of robustness, interpretability, and high performance on structured and text-derived features. For capturing advanced contextual relationships in email text, BERT is employed so as to provide better classification of complicated or ambiguous emails. VADER was picked for sentiment analysis because it is, out of the box, optimized for short, informal text and provides fast, rule-based sentiment scoring suitable for real-time email processing.

Database

MongoDB will be used as the major database for storing raw emails, metadata, and other information that's fed in from various sources like models' outputs and feedback from users or generated analytics logs. MongoDB's document-based nature fits well with the semi-structured nature of the information in emails.

Frontend and Visualisation

React is used in creating an interactive web-based email client/user interface along with an analytics dashboard for real-time visualization of classification results or output metrics.

API and Integration

Gmail API is used for the secure access of the email of the user with specific permission. A RESTful API will act as a link between the backend of the machine learning model and the frontend.

Project Timeline

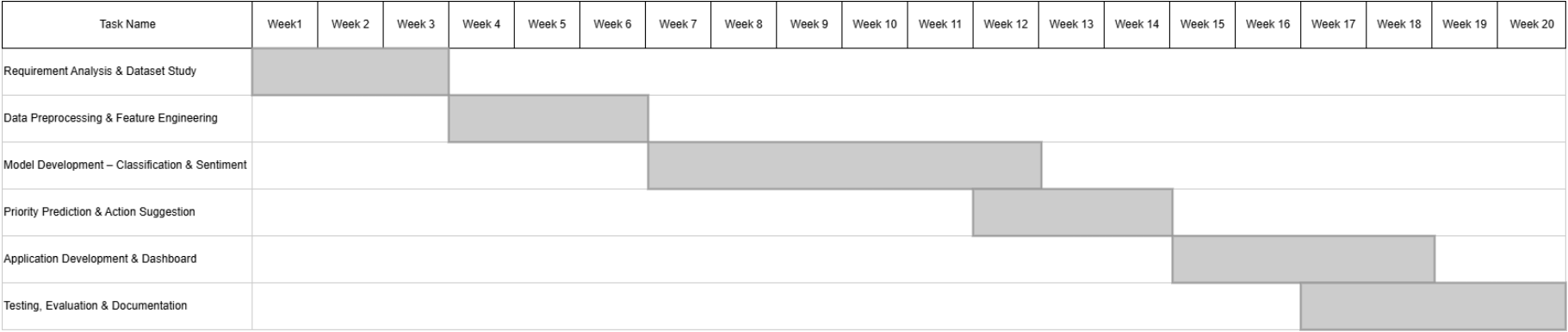


Figure 2: Gant Diagram

Step	Weeks	Task Description
1. Requirement Analysis & Dataset Study	1–3	Define project scope and objectives, review related literature, and explore the Enron and SpamAssassin datasets
2. Data Preprocessing & Feature Engineering	4–6	Clean and normalize email data, extract metadata, and generate textual and statistical features for machine learning
3. Model Development – Classification & Sentiment	7–12	Train and validate email classification models and implement sentiment analysis
4. Priority Prediction & Action Suggestion	13-14	Design priority scoring mechanism and map predicted priorities to suggested user actions
5. Application Development & Dashboard	15-17	Develop the email inbox interface and analytics dashboard, and integrate machine learning backend services
6. Testing, Evaluation & Documentation	18-20	Perform system testing and evaluation, analyze model performance, finalize the application, and prepare project documentation and technical paper

Table 1: Project Timeline

Expected Outcomes

The expected outcomes to be obtained from the proposed Smart Email Management System will include a well-functioning model capable of classification, prediction, analysis, and suggestion. It will include:

- Accurately classify emails into work, personal, spam, and urgent categories using a variety of machine learning strategies on the Enron dataset and SpamAssassin dataset.
- Analyze the tone and sentiment in the received emails and suggest action items such as reply, archive, or flag.
- Include a responsive web interface with an analytics interface for users to observe their patterns of communication, emails, and productivity statistics.

Overall, the framework has the following goals: increase in productivity, email overload reduction, and the ability to make data-driven decisions. The project will also show the application of the machine learning algorithm to real-world email data.

Conclusion

The project proposition will cover the design of Smart Email Management System that features classification, sentiment analysis, and priority prediction to increase the productivity and manageability of user emails effectively. The system will be able to make use of existing data sets such as the Enron data set and SpamAssassin data set by blending robust machine learning techniques to extract significant information and analytics through a flexible user interface on the World Wide Web. The project can be achieved within 18 weeks since it tackles a real-world communication issue by applying data mining and machine learning techniques effectively.

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